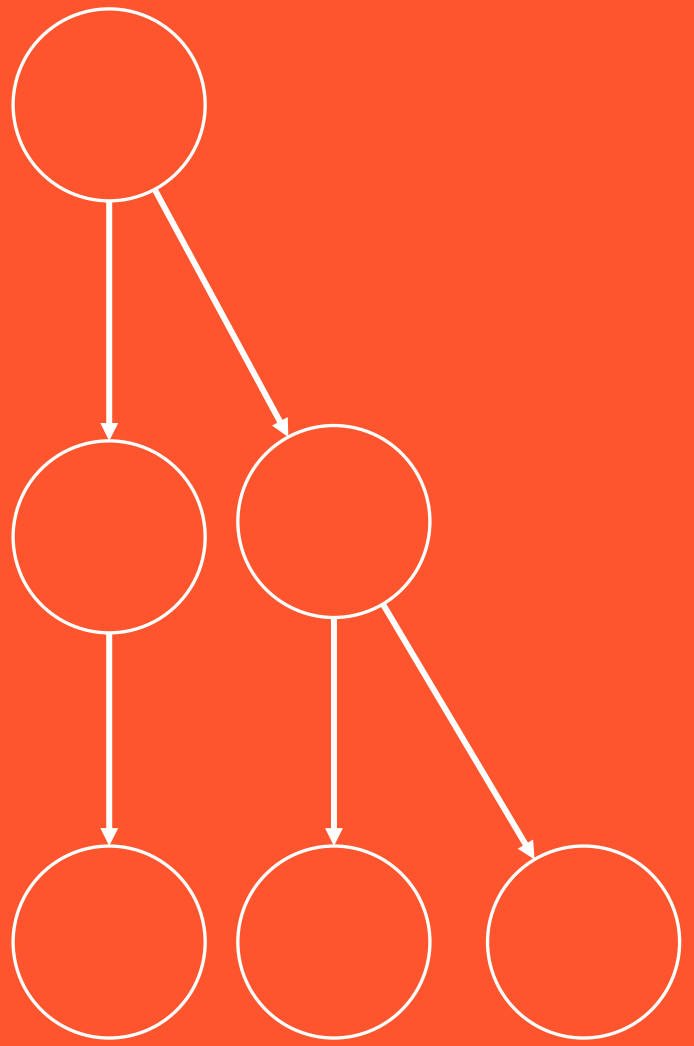


BCOG 100

Learning and Plasticity

Lecture 5.2 — Algorithms for Changing Neural Weights

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

Hebb's Rule: Neurons That Fire Together, Wire Together

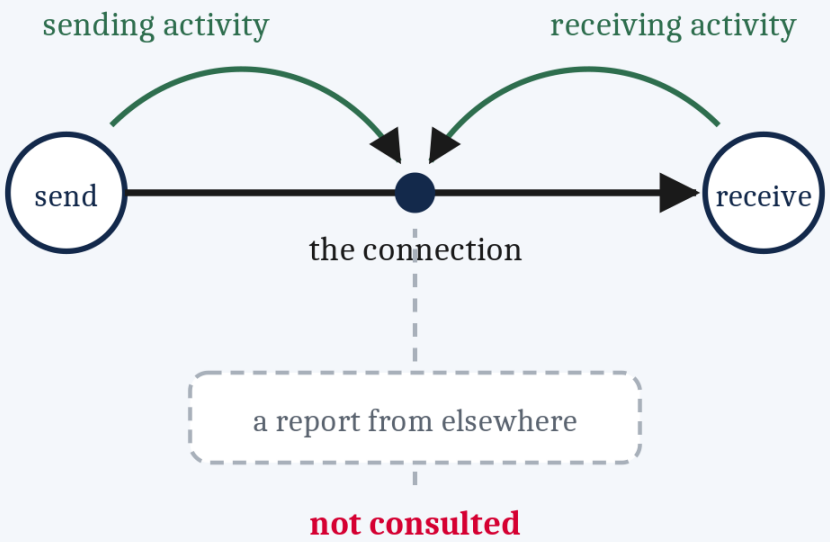
Hebb's rule: $\Delta b_1 = \eta x_1 y$. The change in a weight is the product of the two cells' firing rates, scaled by the **learning rate** η.

Learning rate η	Input rate x_1	Receiving rate y	Change Δb_1
1	40	40	1600
1	40	2	80
1	2	2	4
0.01	40	40	16

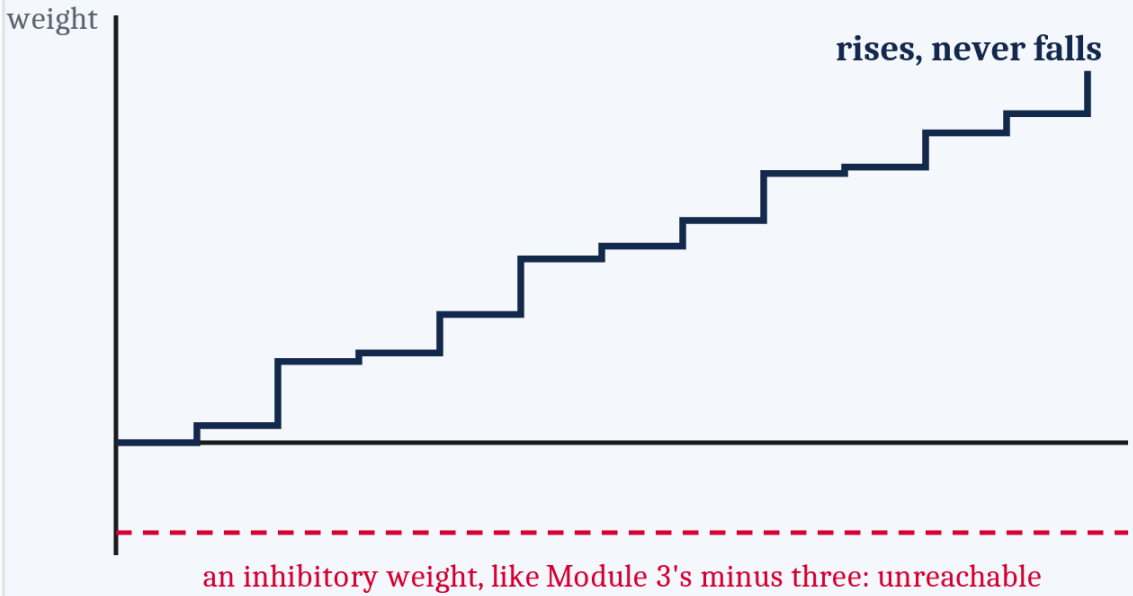
Every Term Is Local, and Every Change Is Upward

Firing rates are never negative, and neither is η, so Δb_1 never is. A Hebbian weight can grow without limit and can never shrink. And everything the rule uses is already at the connection.

Only what is at the connection



The weight across twelve occasions

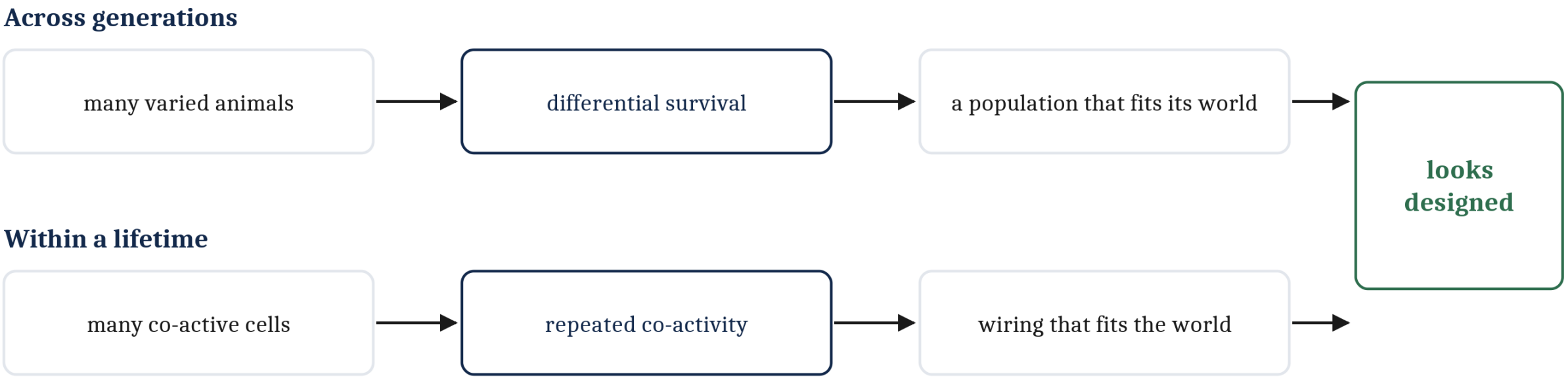


Firing rates are never negative, and neither is the learning rate.

So every change this rule makes is upward.

Coincidence Without a Goal

Nothing in Hebb's rule represents a goal or registers being wrong. The wiring ends up fitted to the world anyway: the shape of selection, one level down and on a far shorter timescale.

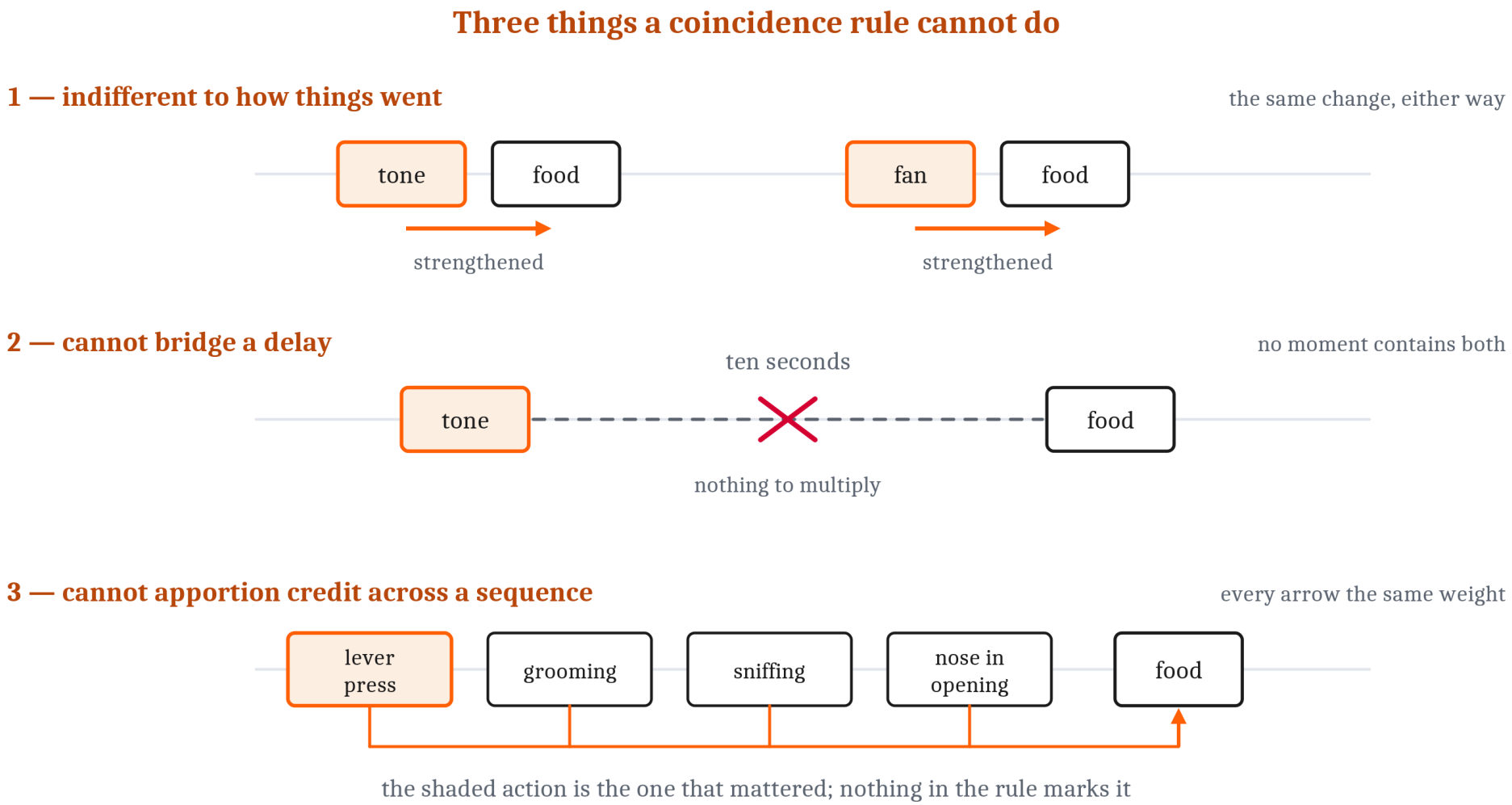


No goal, no comparison, no foresight -- in either row.

The fit is what is left over, not what anything was aiming at.

Three Things Coincidence Cannot Do

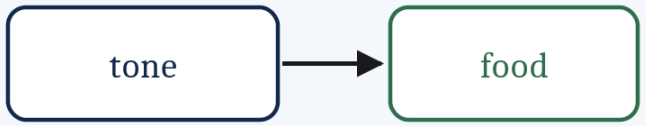
Hebb's rule is **indifferent to outcomes**, it **cannot bridge a delay**, and it **cannot apportion credit** across a sequence. Each one blocks something the first lecture said a learner needs.



Blocking: Pairing Is Not the Point

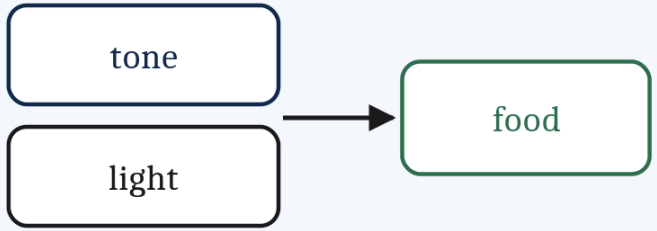
Light and food perfectly paired, and the rat learns almost nothing about the light. **Blocking**: learning tracks what was not already predicted, not what merely co-occurred.

Phase 1: many trials



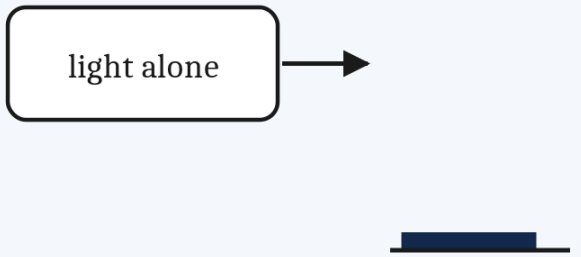
the tone comes to predict the food

Phase 2: many trials



every light is followed by food, and no food arrives without one

The test



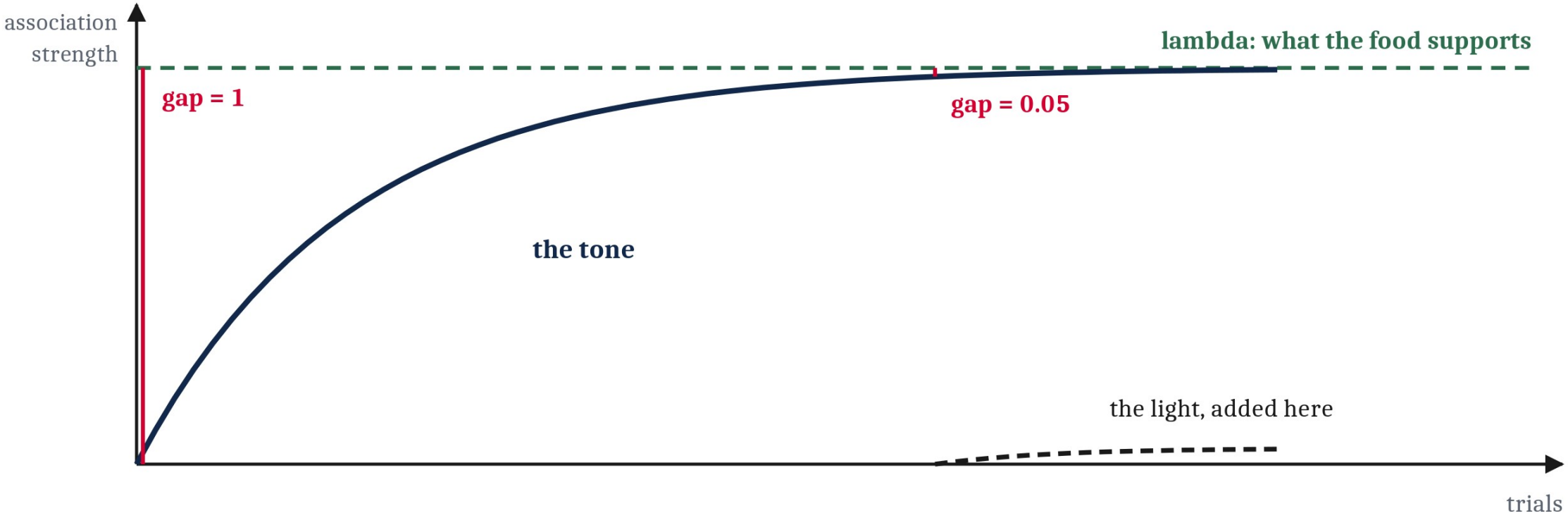
response: **almost nothing**

The food was already predicted, so nothing about the light was news.

Blocking is well documented in mammals; the invertebrate case is unsettled.

Rescorla and Wagner: Learn in Proportion to the Surprise

$\Delta V = \alpha\beta(\lambda - V)$. V is what the cue currently predicts; λ is what the outcome supports. Their difference is the **prediction error**, and when it reaches zero, nothing is left to learn.



The light arrives when there is almost nothing left to learn.

The Same Rule, Discovered Twice

The **delta rule**: $\Delta b_1 = \eta(t - o)x_1$. Hebb multiplies the input by the receiving cell's activity; the delta rule multiplies it by how wrong the receiving cell was.

One rule, written down twice

Rescorla and Wagner, 1972

the animal

$$\Delta V = \alpha\beta(\lambda - V)$$

prediction error

rate terms | what the outcome will support | the strength already present

the same three terms

Widrow and Hoff, 1960

the circuit

$$\Delta b_1 = \eta(t - o)x_1$$

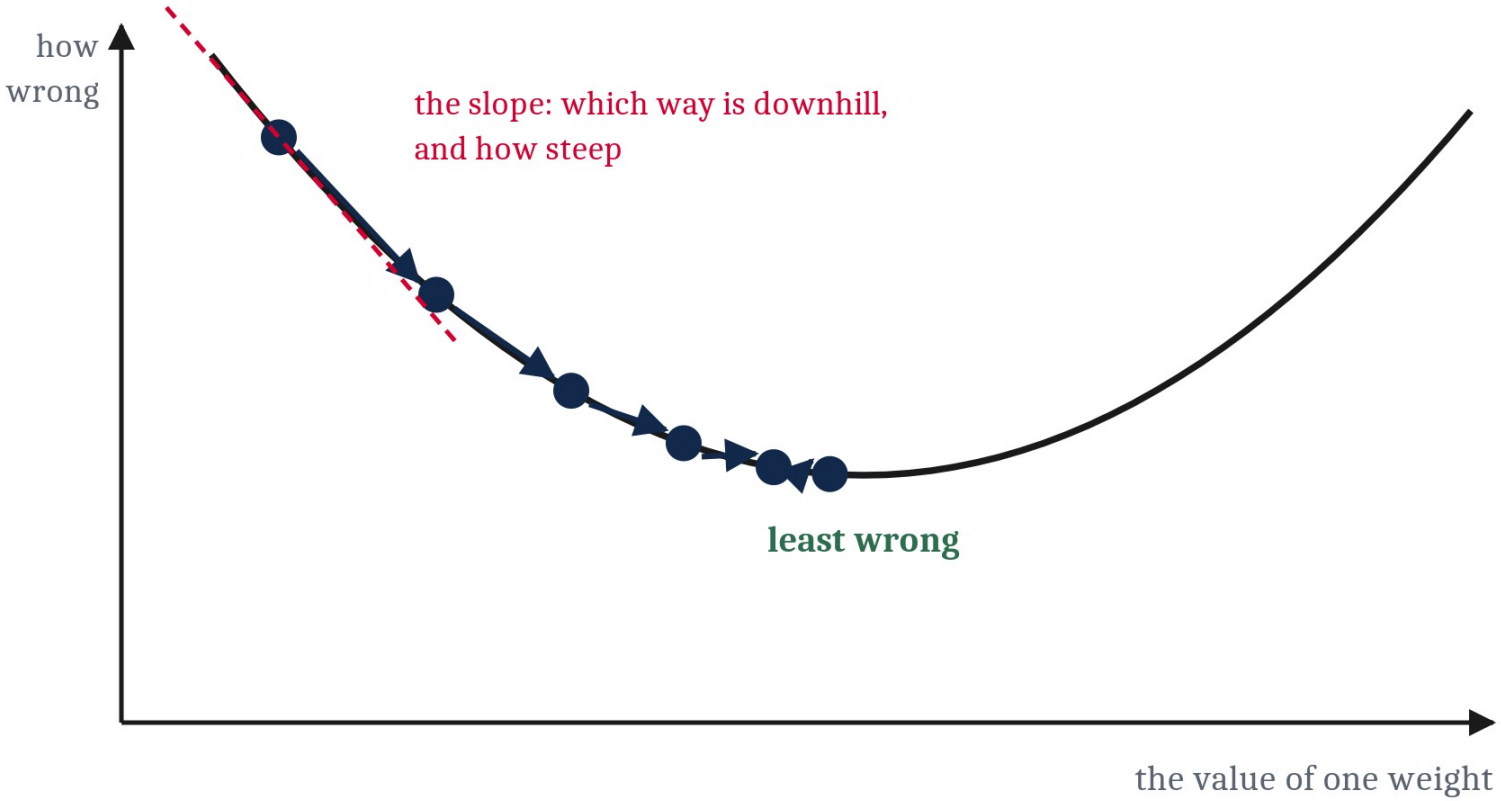
learning rate | target | actual output | input

prediction error

the input term is 1 or 0 on a conditioning trial, which is why the animal version leaves it out

When "It Was Wrong and Corrected Itself" Is Literally True

The delta rule holds an expectation and computes a difference, so error language describes the mechanism. **Gradient descent** is the general method: find which way is downhill, and step.



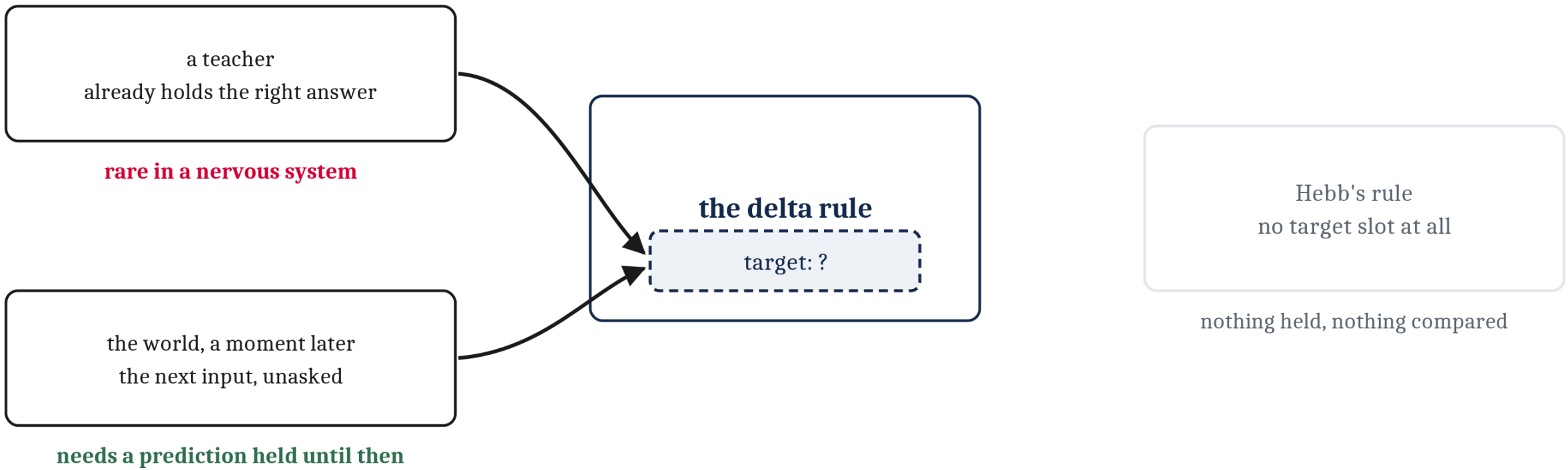
For a single unit, this is exactly the delta rule.

A population on an adaptive landscape computes no slope. The climb is what differential survival leaves behind.

Error language earns its place where the mechanism holds an expectation and computes a difference.

Where the Target Comes From

The delta rule needs a target. A teacher can supply one, and nervous systems have few. Or the target can be the next thing that happens: **prediction learning**, which holds a prediction until the answer arrives.



The rule is the same rule; only the supplier of the target changes.

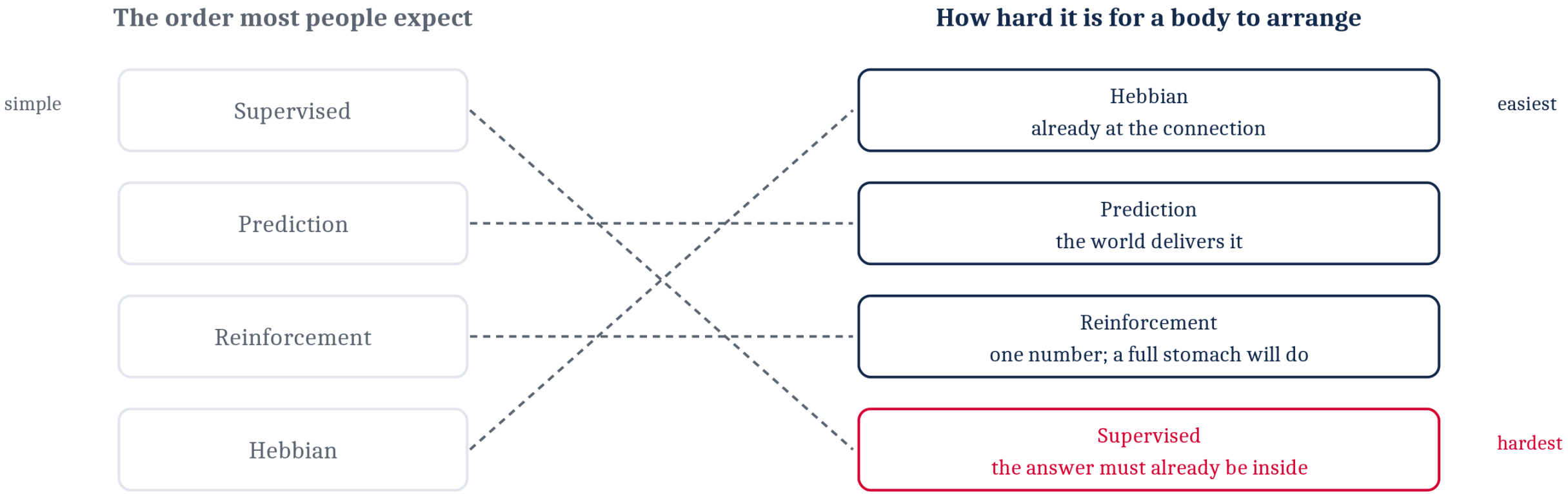
Four Kinds of Learning, Sorted by What a Weight Change Depends On

The four rules differ in one respect: what information has to be present for a weight to move. They are four situations an animal can be in, and nothing makes them a sequence.

Kind of learning	What the weight change depends on	Where that information comes from	How readily a nervous system could supply it
Hebbian (unsupervised)	The activity of the two cells a connection joins	Nowhere else; both quantities are already at the connection	Trivially — nothing has to be arranged
Prediction (self-supervised)	The difference between what was predicted and what then arrived	The world, which delivers the answer a moment later	Easily, provided a prediction can be held until the input arrives
Supervised	The difference between the output produced and the output specified	A teacher that already holds the correct answer	With difficulty; the answer has to exist somewhere inside the animal already
Reinforcement	One number reporting how things went	A verdict generated by the body or by the outcome itself	Readily — one broadcast quantity is enough

Ranked by What a Body Can Arrange, the Order Flips

A teacher is the easiest kind of learning to write down and the hardest for a body to arrange. In a nervous system, supervised learning is the exotic case.

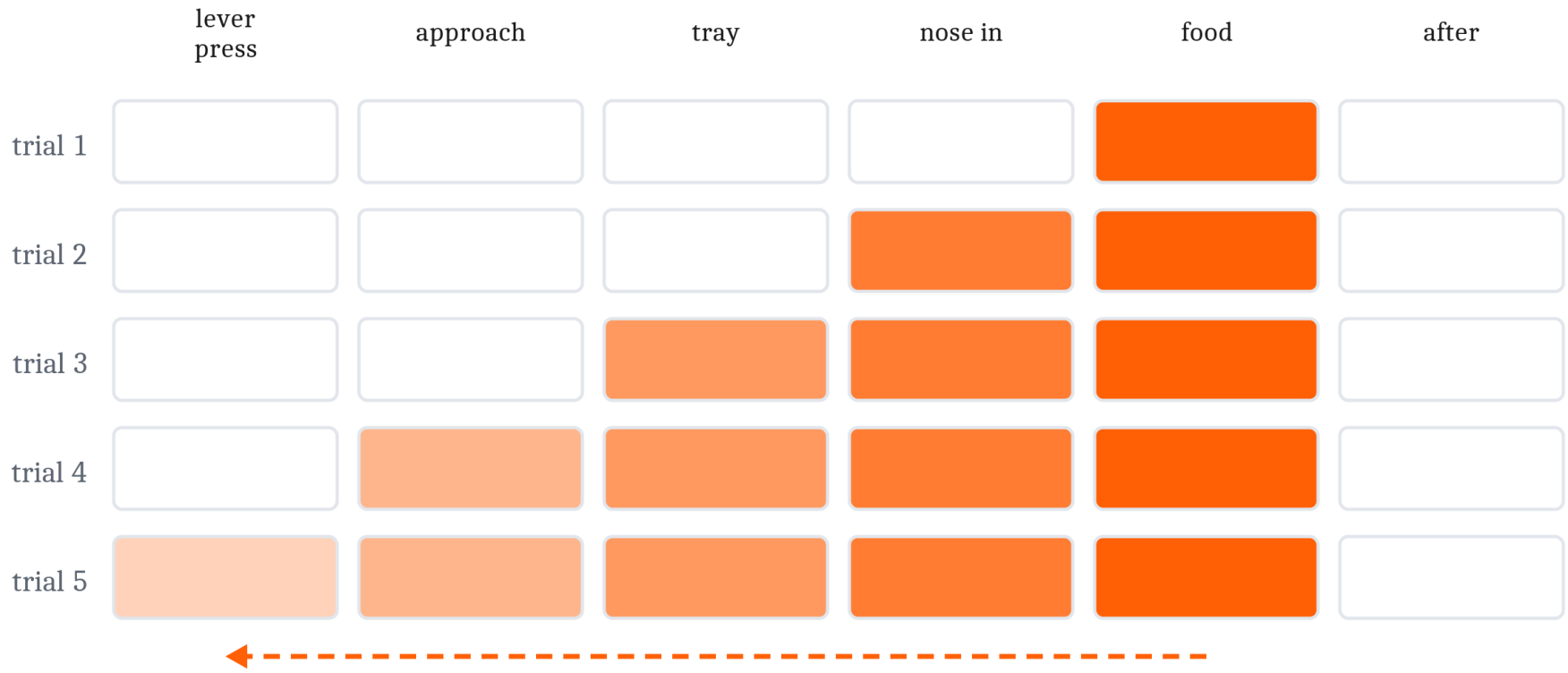


Easiest to write down, hardest to arrange: in a nervous system supervised learning is the exotic case.

Bridging the Delay: Learn From the Next Estimate

Discounting makes a distant outcome count for less. **Bootstrapping** compares each estimate against the next one, so an error can move a weight right away — **temporal-difference learning**.

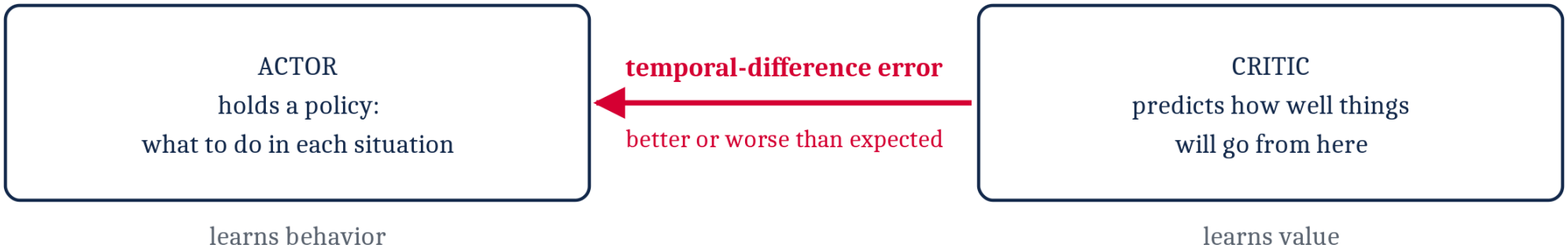
Value seeps backward along the chain, one trial at a time

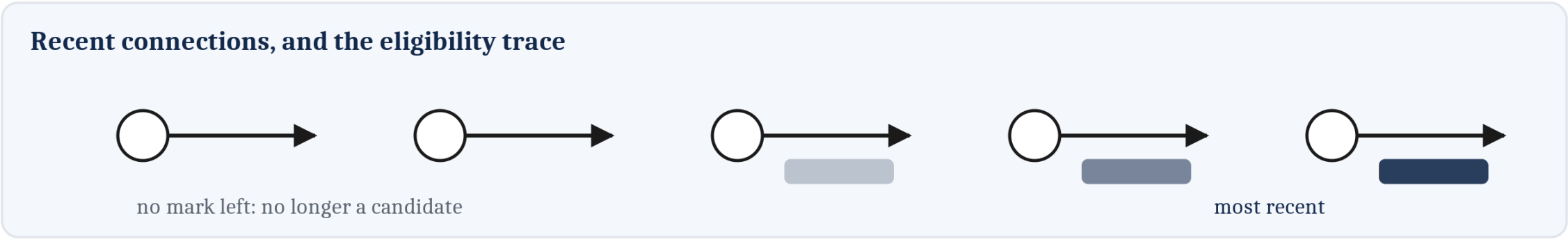


by the last trial the lever press carries it, and the food itself is no longer a surprise

Animals Bootstrap, and Judging Can Be Separated From Choosing

The **critic** learns **value**; the **actor** holds a **policy** and adjusts it by the critic's errors — an **actor–critic** architecture. An **eligibility trace** marks which connections were recently involved.



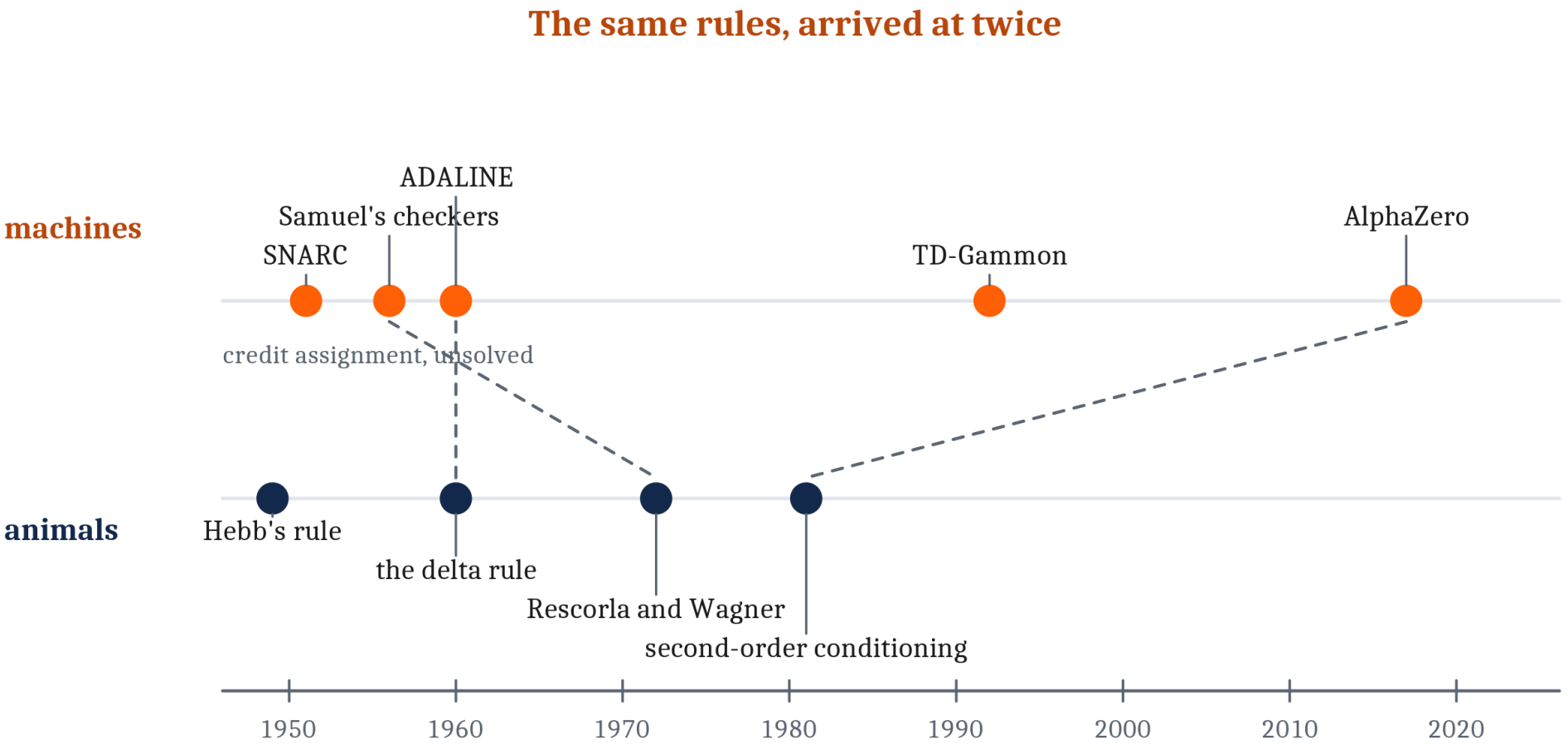


A later signal finds the connections that still carry a mark.

The trace is a proposal about a procedure, not yet a claim about tissue.

The Same Rules, Built in Machines

ADALINE ran the delta rule in 1960. SNARC hit the credit assignment problem in 1951. Samuel's checkers program ran temporal-difference learning. AlphaZero is actor and critic, made very large.

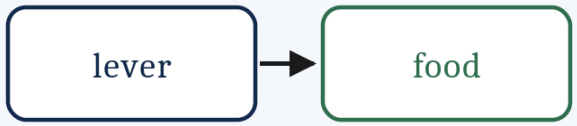


dashed lines join the pairs that are the same algorithm

Habits and Goals

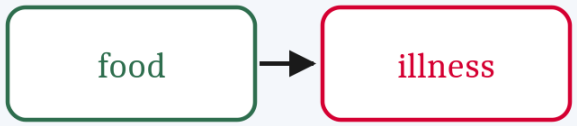
Model-free: repeat what has worked here. **Model-based**: run a model of what leads to what. Devaluation tells them apart, and extended training turns **goal-directed** behavior into a **habit**.

1. Train



press for a
particular food

2. Away from the lever



until the rat
refuses the food

3. Back at the lever

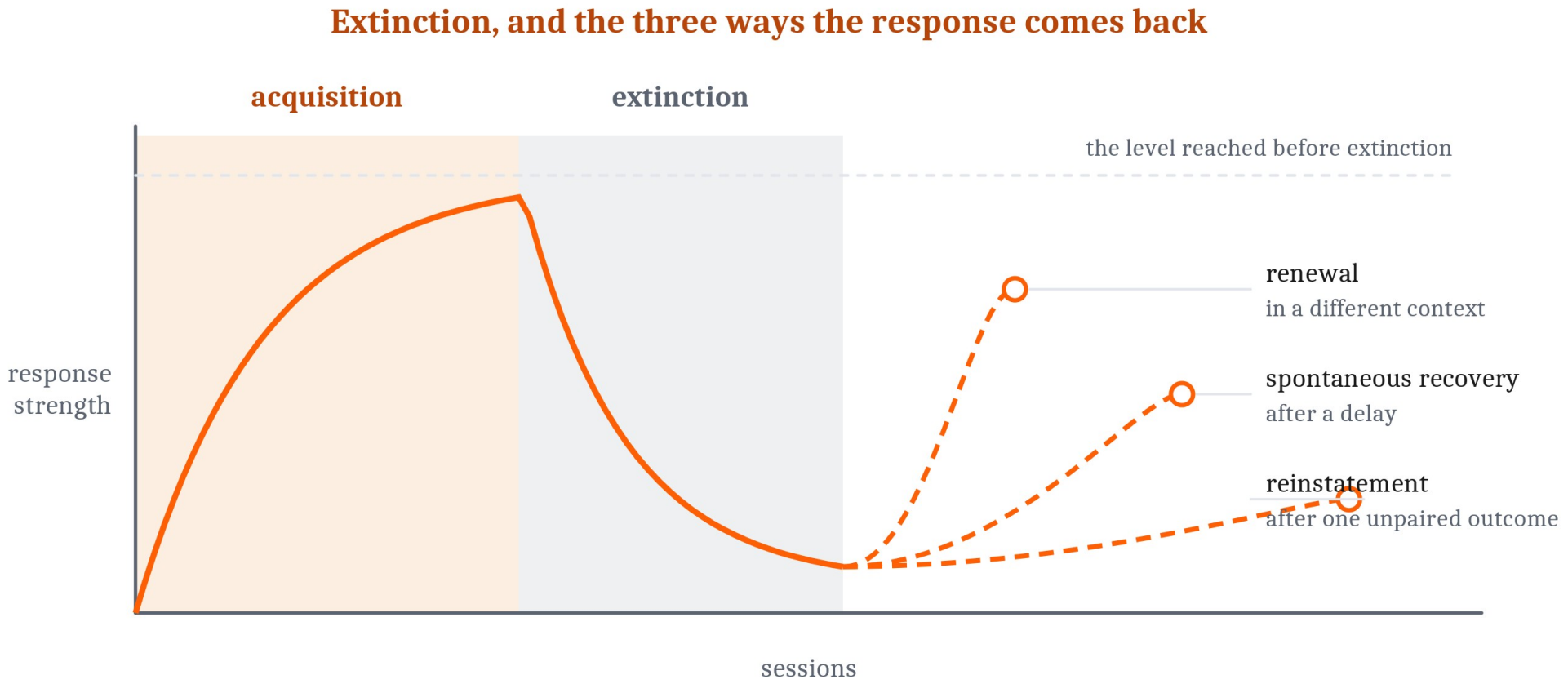
modest training stops pressing	extended training keeps pressing
GOAL-DIRECTED	**HABIT**
the model says the lever leads to unwanted food	nothing about the lever itself has changed

Model-based choice stops at once. Model-free choice keeps going.

Which one appears depends on how much training came first.

Extinction Is Not Unlearning

Spontaneous recovery, the **renewal effect** and **reinstatement** all bring an extinguished response back, which a weight that had simply gone down could not. **Extinction** is new learning.



a weight returned to its starting value could produce none of this

What Cannot Be Learned, and Why That Is Not a Flaw

The rules have no preferences. The animal does, and they line up with the pairings its ancestors had reason to pick up — an **inductive bias**.

Paired with	Illness	Electric shock
A novel flavor	Learned in one pairing, across hours	Hardly learned at all
A noise	Nearly hopeless	Learned readily

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